MYCN-Adjusted Falsification and ERF-Style Evidence Receipt Documentation of Immune-Evasion Hypotheses in Pediatric Neuroblastoma

ImmuneErrorRadar falsification protocol with ERF-style receipts, calibration ledger, and three-cohort / cross-tumor expansion guard

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Preprint status and claim boundary: not peer reviewed. This work reports computational hypothesis triage, evidence-readiness review, and falsification receipts only. It makes no diagnostic, therapeutic, clinical, wet-laboratory efficacy, or biological-proof claim.

Abstract

Background: Bulk transcriptomic studies of immune evasion in pediatric neuroblastoma can report gene-expression signals that are confounded by MYCN amplification. At the same time, AI-assisted hypothesis generation increases the need for pre-laboratory filters that can kill weak or mis-specified claims before experimental resources are spent.

Methods: We developed ImmuneErrorRadar v3.0.6.9.9, a falsification-first hypothesis stress-testing and investment-triage framework with Vraimony/ERF-inspired documentation inside the methods layer. The current baseline adds an ERF-inspired report receipt discipline, hypothesis-derived report titles, report history, a calibration ledger, and a three-cohort / cross-tumor expansion guard. Candidate immune-evasion axes were reviewed in GSE85047 using event-rate screening, Cox proportional hazards models, MYCN-adjusted Cox models, MYCN-stratified summaries, component separation, targeted GPL5175 probe-resolution receipts, SHA-256 reproducibility receipts, evidence-readiness tiers, and a kill/data-only decision layer.

Results: MYCN adjustment eliminated or substantially attenuated several raw immune-survival signals. CK2/CSNK2A1 and HDAC/DNMT upstream mechanisms were killed as written because the local APM signal could not be attributed to those mechanisms from static bulk expression. NECTIN2-TIGIT was killed as a bulk-expression testability claim, not as a biological disproof. Gamma-delta/NKG2D remained surrogate-only pending TRGC/TRDV/TCR-aware evidence. The APM/B2M-TAP-HLA axis was the only convergent downstream direction retained for data investment, not wet-laboratory investment.

Conclusions: ImmuneErrorRadar converts evidence-readiness into explicit research-investment decisions: DO_NOT_INVEST_CLAIM_KILLED, WATCHLIST_ONLY, NOT_MEASURABLE, or INVEST_IN_DATA_ONLY. In this Tier-0 analysis, no hypothesis reached wet-laboratory candidacy. The principal output is not proof of an immune-evasion mechanism, but a reproducible, claim-bounded falsification record showing which claims should be killed as written and which, if any, warrant additional data acquisition and independent replication.

Keywords: neuroblastoma; MYCN amplification; antigen presentation; immune evasion; computational falsification; Vraimony; ERF-style evidence receipts; investment triage; evidence readiness; Cox regression; APM; calibration ledger; AI-generated hypotheses.

1. Introduction

Pediatric neuroblastoma is a clinically heterogeneous childhood cancer in which MYCN amplification remains one of the most important adverse molecular features. Because MYCN amplification reshapes tumor biology and the tumor immune microenvironment, immune-related expression signatures derived from bulk RNA data can be confounded if MYCN is not explicitly modeled.

The current research environment also faces a second pressure: hypothesis generation is becoming easier than hypothesis validation. AI systems can generate and rank mechanistic cancer hypotheses at scale. This is valuable, but it also increases the risk that weakly grounded hypotheses enter expensive wet-laboratory pipelines without adequate falsification.

This study presents ImmuneErrorRadar as a falsification-first framework. The aim is not to prove biology, diagnose disease, or nominate therapy. The aim is narrower and more operational: given a supplied computational immune-evasion hypothesis, determine whether the claim should be killed as written, placed on a watchlist, rejected as not measurable, or advanced only to further data acquisition before any laboratory investment.

The central design choice is deliberately conservative. A downstream expression-survival direction is not allowed to validate an upstream mechanism. A bulk-expression proxy is not allowed to validate ligand-receptor engagement. A local raw signal is not allowed to survive as an investment candidate if it is erased or strongly attenuated after MYCN adjustment. The framework separates directional signal, mechanistic attribution, platform/probe readiness, replication readiness, cross-cohort scope, documentation integrity, and research-investment decision.

2. Methods

2.1 Protocol identity and Vraimony-powered report standard

Analyses were consolidated against ImmuneErrorRadar v3.0.6.9.9, whose report baseline is “Hypothesis Stress-Test & Investment Decision Report.” The methodology block documents: Protocol: ImmuneErrorRadar falsification protocol; Documentation standard: ERF-inspired evidence receipt discipline; Powered by Vraimony as a methods-level provenance note, not as a biological or clinical claim.

The protocol is designed as a followed methodology, not a guarantee of truth. Its function is to create a bounded falsification record: what claim was supplied, how it was routed, what was tested, what was not measurable, which gates blocked promotion, and what decision resulted.

2.2 ERF-inspired evidence receipt discipline

The report receipt discipline is inspired by Vraimony-style ERF thinking: hash-first, portable, replayable, reviewer-readable, and claim-bounded. A report receipt can document the payload hash, replay identity, protocol name, decision state, and missing gates. It does not prove source truth, clinical validity, or biological mechanism.

The current Android-local build produces local falsification receipts only. Unless future Vraimony server signing is implemented, reports should state signedByVraimony=false. This distinction prevents confusion between a local methodology receipt and a server-signed certificate.

2.3 Dataset and scope

The primary dataset was GSE85047, a pediatric neuroblastoma expression cohort generated on the Affymetrix Human Exon 1.0 ST array platform (GPL5175). The analysis used the subset with progression-free survival information and MYCN amplification status available for in-app survival screening.

The primary outcome was progression-free survival. MYCN amplification was treated as a binary covariate. MYCN-stratified summaries were generated for amplified and non-amplified subgroups. Independent time-to-event replication remains pending; therefore, no Tier-2 computational evidence claim is made.

2.4 Three-cohort / cross-tumor expansion guard

Because a single neuroblastoma cohort is not enough to support a broad pediatric oncology claim, v3.0.6.9.8 uses a three-layer validation guard. The first layer is the primary neuroblastoma stress-test cohort (GSE85047). The second layer is neuroblastoma replication/data-acquisition, with GSE49710 and TARGET-NBL treated as required replication candidates before Tier-2 language. The third layer is pediatric tumor-family expansion: medulloblastoma, Wilms tumor/nephroblastoma, and Ewing sarcoma.

This guard is not executed replication. It is a scope-control and data-fuel mechanism. It prevents pan-pediatric-tumor or wet-laboratory language from being promoted from neuroblastoma-only evidence. The current state is NEUROBLASTOMA_ONLY_SINGLE_COHORT_SCOPE with crossTumorPromotionAllowed=false and wetLabPromotionAllowed=false.

2.5 Probe-resolution and vector-binding policy

Gene-to-probe handling is separated into platform mapping and expression-vector binding. The framework records whether a gene is present in the targeted GPL5175 review manifest, whether the candidate probe is present in the matrix subset, whether the probe is absent from the platform, and whether multi-probe ambiguity prevents automatic selection.

The current analysis uses a targeted static GPL5175 review manifest and legacy mapping support as review-only evidence. Probe resolution is not treated as certified source truth until full GPL5175 annotation certification is completed. A gene can be mappable on the platform while still lacking a bound expression vector in the current matrix subset.

2.6 Statistical workflow and evidence tiers

For executable axes, the framework ran event-rate screening, univariate Cox proportional hazards models, and MYCN-adjusted Cox models. Where appropriate, component separation compared event-rate deltas for subcomponents such as antigen-presentation loss versus NK-effector loss. MYCN attenuation is interpreted conservatively: a raw association that weakens after MYCN adjustment is not eligible for mechanistic or wet-laboratory investment without independent replication.

Tier	Requirement	Allowed claim
Tier-0	Slice reproducibility, local run receipt, review-only probe state	Directional screening and kill/data-only triage only
Tier-1	Certified platform/probe annotation and reproducible expression-vector binding	Stronger directional signal claim; no biology proof
Tier-2	Independent cohort replication with time-to-event Cox and stable direction	Computational evidence claim; still not clinical or therapeutic proof

All results reported here remain Tier-0. No Tier-1 or Tier-2 claim is made.

2.7 Investment-triage and calibration ledger

The investment-triage layer converts evidence-readiness results into explicit research decisions: DO_NOT_INVEST_CLAIM_KILLED, WATCHLIST_ONLY, NOT_MEASURABLE, NULL_LIKE_OR_INCONCLUSIVE, SURVIVED_LOCAL_DIRECTION_REVIEW_ONLY, or INVEST_IN_DATA_ONLY. It does not report numeric translation probabilities because no calibrated dataset yet links these specific computational criteria to future laboratory success.

Every tested hypothesis is also stored as a calibration-set row. The row records a pattern outcome such as killed, data-only survivor, watchlist, not measurable, null-like, or review-only survivor. This creates a historical translation memory: over time, the platform learns which claim patterns usually collapse, survive as data-only, or remain unmeasurable.

2.8 Hypotheses evaluated

The framework stress-tested executable immune-evasion axes including hypoxia/stroma, CD276/B7-H3, mesenchymal/stromal program, ODC1/polyamine, APM/B2M-TAP-HLA, gamma-delta/NKG2D surrogate, CK2/APM fallback, HDAC/DNMT/APM fallback, NK-effector activity, and NECTIN2/TIGIT checkpoint testability. NECTIN2/TIGIT was treated as a checkpoint claim that is not testable from bulk expression without single-cell or functional ligand-receptor data.

3. Results

3.1 Summary of final decisions

Claim/axis	Final decision	Rationale	Allowed next investment
CK2/CSNK2A1 mechanism	DO_NOT_INVEST_CLAIM_KILLED	APM signal exists but cannot be attributed to CK2; raw signal weakens after MYCN; no perturbation/protein/IFN-context mechanism evidence.	None for CK2; mechanism-agnostic APM data only.
HDAC/DNMT mechanism	DO_NOT_INVEST_CLAIM_KI	APM signal exists but cannot	None for HDAC/DNMT; APM

	LLED	be attributed to epigenetic repression; no chromatin, methylation, or perturbation data.	data replication only.
NECTIN2/TIGIT bulk checkpoint	DO_NOT_INVEST_CLAIM_KILLED for current data	Bulk selected matrix cannot test ligand-receptor checkpoint function; no executable local Cox model.	Reopen only with single-cell/functional checkpoint data.
Gamma-delta/NKG2D	WATCHLIST_ONLY / surrogate not true gamma-delta evidence	TRGC/TRDV/TCR identity is not directly measured; surrogate direction is adverse and MYCN-attenuated.	Only with TRGC/TRDV/TCR-aware mapping or single-cell data.
ODC1/polyamine	WATCHLIST_ONLY / primary-driver claim killed	Weak secondary signal; no robust independent MYCN-adjusted support.	No primary investment.
CD276/B7-H3 pooled effect	WATCHLIST_ONLY / pooled main effect killed	Near-null pooled MYCN-adjusted effect; possible subgroup direction underpowered.	Only a dedicated MYCN-amplified subgroup study.
Hypoxia/stroma	DO_NOT_INVEST_CLAIM_KILLED	Signal is absorbed by MYCN structure or cellular-mixture confounding.	No independent PFS investment from this route.
Mesenchymal/stromal counter-signal	WATCHLIST_METHOD_COUNTERSIGNAL_NO_INVEST	Counter-signal likely reflects bulk cellular mixture or tumor-purity effects, not tumor-intrinsic mesenchymal immune exclusion.	Deconvolution or single-cell support only.
APM/B2M-TAP-HLA	INVEST_IN_DATA_ONLY	Convergent downstream axis across CK2, HDAC and direct-expression routes; not strong enough for wet-lab because MYCN-adjusted signal is attenuated.	Full GPL5175 mapping, vector extraction, GSE49710/TARGET-NBL replication, then cross-tumor expansion.

3.2 MYCN-shadow attenuation pattern

The dominant pattern was attenuation of raw immune-survival associations after MYCN adjustment. The most important interpretation is not that all immune biology is false; rather, claims of independent prognostic or mechanistic activity cannot be defended when the signal weakens after MYCN modeling.

Axis	**Raw result**	**MYCN-adjusted result**	**Decision**
Hypoxia/stroma proxy	raw p approximately 0.005	eliminated after MYCN stratification	Killed as independent PFS signal
CK2/APM fallback (N=272, events=96)	HR=1.416, p=0.012 (N=272, events=96)	HR=1.197, p=0.246 (N=272, events=96)	CK2 mechanism killed; APM local direction survives only as data target
Gamma-delta/NKG2D surrogate (N=272, events=96)	HR=1.428, p=0.010 (N=272, events=96)	HR=1.223, p=0.207 (N=272, events=96)	True gamma-delta claim not supported; surrogate only
B2M/APM/NK decoupling (N=272, events=96)	HR=1.438, p=0.006 (N=272, events=96)	HR=1.231, p=0.183 (N=272, events=96)	Independent MYCN-adjusted claim killed; data follow-up allowed due convergence

Table note: Executed Cox rows shown for CK2/APM fallback, gamma-delta/NKG2D surrogate, and B2M/APM/NK decoupling use N=272 and events=96 in the current Tier-0 selected-matrix review output. Hypoxia/stroma is reported as a route-specific attenuation summary.

3.3 APM/B2M-TAP-HLA is retained only as a data-investment candidate

The antigen-presentation machinery axis was the only downstream direction retained for further data investment. This decision is not based on definitive MYCN-adjusted survival proof. It is based on convergence: several mechanistically distinct upstream hypotheses collapsed onto the same downstream APM-low phenotype.

Because the MYCN-adjusted signal remains attenuated and independent replication is absent, the correct decision is INVEST_IN_DATA_ONLY. The allowed next work is full GPL5175 probe/vector completion, independent cohort replication, direction-stability testing, and bulk-attribution/deconvolution checks. Wet-laboratory experiments are not justified from the current evidence.

3.4 Mechanism-specific claims killed as written

CK2 and HDAC/DNMT hypotheses produced APM fallback signals but did not produce direct evidence for the upstream mechanisms. Static bulk expression cannot test silmitasertib-interferon synergy, phosphorylation or destabilisation of B2M/HLA-A, chromatin state, DNMT methylation, or perturbation response. Therefore the correct decision is DO_NOT_INVEST_CLAIM_KILLED for the supplied upstream mechanism.

NECTIN2-TIGIT illustrates a different disposition. The framework does not claim that single-cell NECTIN2-TIGIT biology is false. It kills the claim that bulk GSE85047 selected expression can test ligand-receptor checkpoint function. The allowed next investment is not more bulk proxy analysis; it is a single-cell or functional checkpoint dataset designed to measure ligand-receptor context.

The gamma-delta/NKG2D route remained surrogate-only. Because TRGC/TRDV/TCR identity is not directly measured by the current selected matrix, the result cannot be interpreted as true gamma-delta T-cell evidence.

3.5 Vraimony receipts and report history

The current report format can include a methods-level “Powered by Vraimony” provenance block and an ImmuneErrorRadar falsification protocol line. Each report can include a falsification receipt identifier and content hash. The receipt documents workflow and payload integrity only; it does not certify biological truth or source-truth validity.

The user interface now stores reports by date and derives titles from the hypothesis itself. This converts report history from a generic log into an evidence-review library. The copy, download and print actions allow a researcher to export selected reports for review.

3.6 Calibration ledger and translation memory

The calibration ledger is the foundation for future translation. Each tested hypothesis adds a row recording whether the pattern was killed, survived only as data-only, remained watchlist, became not measurable, or appeared null-like. This does not convert the current analysis into a probability model. It creates the infrastructure for future calibration once enough labeled outcomes exist.

4. Discussion

The main contribution of this work is the separation of computational signal from investment decision. Standard transcriptomic workflows often end with “promising but requiring validation.” In a low-translation environment, that conclusion is operationally weak. ImmuneErrorRadar instead asks whether the claim as written deserves additional research spending.

The strongest scientific pattern is MYCN-shadow attenuation. Several raw immune-associated survival signals weaken after MYCN adjustment, supporting the concern that MYCN amplification can create the appearance of independent immune-evasion biomarkers in bulk expression data. This finding justifies requiring MYCN adjustment before any neuroblastoma immune-survival claim is advanced.

The most important positive output is the APM convergence result. APM is not presented as proven biology or as a wet-laboratory target. It is presented as the only downstream axis that deserves more data because multiple killed upstream hypotheses converged on it. This is a narrow decision: invest in mapping and replication, not mechanism-specific experiments.

The Vraimony-powered documentation layer strengthens the manuscript by making the protocol auditable. Instead of asking readers to trust the narrative, the report provides receipts, hashes, boundaries, missing gates, and decision states. This is a documentation advance, not a biological proof claim.

The three-cohort / cross-tumor guard is essential for credibility. Neuroblastoma-only evidence remains disease-scoped. Cross-pediatric-tumor credibility requires independent human datasets from additional pediatric tumor families before any generalization, Tier-2 promotion, or wet-laboratory language.

5. Limitations

This is a Tier-0 analysis. Probe mapping is review-only until full GPL5175 annotation certification and vector extraction are complete. Several probes are resolved but not yet bound to expression vectors in the current matrix subset.

Independent time-to-event replication was not completed. Directional replication without survival time is insufficient for Tier-2 evidence. GSE49710, TARGET-NBL or another Cox-capable cohort is required before any computational evidence claim.

Cross-tumor expansion was not executed in this manuscript. Medulloblastoma, Wilms tumor/nephroblastoma, and Ewing sarcoma are validation targets and scope guards, not evidence that the APM pattern generalizes.

Bulk expression cannot distinguish tumor-intrinsic expression from stromal, vascular, or immune-cell mixture. Single-cell, deconvolution, or functional validation is required before cell-type or mechanism claims can be advanced.

The investment-triage layer is criteria-based, not probability-calibrated. It deliberately avoids numeric translation probabilities until a calibration dataset of tracked hypothesis outcomes exists.

The current Vraimony receipt is local and unsigned by the Vraimony server. It documents report payload integrity and falsification workflow only.

6. Conclusion

ImmuneErrorRadar v3.0.6.9.8 is a Vraimony-powered falsification-first and investment-triage framework for computational immune-evasion hypotheses in pediatric neuroblastoma. In this Tier-0 analysis, CK2, HDAC/DNMT, NECTIN2-TIGIT bulk checkpoint, ODC1 primary-driver, CD276 pooled-effect, hypoxia/stroma and true gamma-delta claims were not suitable for further investment as written. The only retained direction was APM/B2M-TAP-HLA, and only for additional data acquisition, mapping and replication.

The practical conclusion is intentionally conservative: no hypothesis reached wet-laboratory candidacy. The preprint contribution is the decision framework itself and the demonstration that explicit MYCN adjustment, probe readiness, mechanism blocking, ERF-inspired documentation, calibration logging, and investment triage can prevent weak computational hypotheses from consuming experimental resources.

Acknowledgments and protocol note

The report-documentation layer is based on Vraimony methodology and an ERF-inspired evidence-receipt discipline: hash-first, portable, replayable, and claim-bounded. This provenance note identifies the documentation standard used by the workflow; it does not imply medical certification, clinical validation, or proof of biological mechanism.

Data and code availability

GSE85047 and GSE49710 are publicly available through NCBI GEO. TARGET-NBL is available through NCI/GDC-linked TARGET resources. Code, receipts, targeted GPL5175 review manifests, and replay documentation are intended for public repository release at https://github.com/laminsid/Immune. If the repository is not yet publicly populated at the time of submission, these artifacts should be deposited or linked with the preprint record upon submission, and the manuscript should be read as a methods-and-results preprint with local reproducibility hashes rather than a fully executable public package.

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Conflicts of interest

The author declares no conflicts of interest.

Ethics statement

This study used public, de-identified transcriptomic datasets and did not involve direct human-subject recruitment or intervention by the author.

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